Blender Globe Project and Route Tool (Aiming for grade C/B)

Background

Humans have been mapping parts of our (roughly) globe shaped earth on 2D planes for millenia, and the practice is critical for properly understanding our world in ways our limited, on-the-ground perspective doesn't allow. But, by their very nature, all such maps must distort reality to represent a 3D globe in 2D space, and each kind of map, each projection scheme, makes different tradeoffs in this regard.

A 2D map has many benefits, being easy to alter or create with any image-editing software, and being easy to display without distortion or loss of information on a 2D screen. But for those viewing maps to gain a deeper understanding, or creating maps of their own depicting real world datasets or fictional worlds, having tools at hand to enable that understanding is critical. Tools that show the user the reality which the 2D map aims to represent.

Specifically for this work, tools that allow the user to toggle between looking at the 2D map and the 3D globe it represents, able to move around in 3D to get the full picture of this globe. But more importantly, tools that accurately display routes connecting points and their lengths. Both rhumb line routes, routes easily navigable with a compass by maintaining a constant course relative to north, and orthodromic routes, i.e. the true shortest routes connecting points.

Problem

The aim of this work is to provide these tools that allow for fuller comprehension of 2D maps, with specific focus on distances between locations. The aim is to create tools in blender, using the highly flexible Geometry Nodes system, to do two things. The first being to allow for the projection of any 2D mercator projection onto a 3D globe easily, such that a user can look at a map they are viewing or creating both in the convenient mercator format and the more accurate globe format so they can really understand what they are looking at. This should be relatively simple.

The second part, expected to be larger in scope, is allowing the user to input the coordinates of any two points, and be shown routes connecting those two points, along with the true length of that route. Both on the 2D Mercator projection and the true 3D globe it represents. The user should be able to toggle between the rhumb line route and orthodromic route connecting these two points.

Implementation

To be determined, but the basic projecting from Mercator to globe should be easily managed by turning existing formulas into geometry node systems.

The drawing of rhumb lines will be somewhat easy since the basis is the Mercator, and where rhumb lines just appear as straight lines, so the problem is as easy as drawing the shortest line between two points (with some caveats like the edges of the map looping around to the other edge). Measuring this distance may be more difficult (since the length of the line on the mercator has nothing to do with the length on the actual globe) but there should be formulas for that.

The real problem will likely be figuring out how to draw the orthodromic route. This will take some research, but may involve a reversed Mercator projection, drawing the route on the sphere first and then projecting that onto the Mercator projection.

Evaluation

Relatively simple, the results will be evaluated by taking an existing Mercator projection of the earth and ensuring that:

1. Coordinates correspond to the locations they should (Stockholm is at the earth coordinates of Stockholm, and same with a number of other locations)
2. The routes, both rhumb line and orthodromic, are the correct length, based on external sources and tools for calculating those distances
3. The routes drawn, of both kinds, correspond to the real routes on the surface of the Earth, confirmed both using inspection of the map and globe as well as external sources.

Development Blog

Link to Devblog:
https://docs.google.com/document/d/15oBxvKo4drUms28vUtmy5OiYSwPEtHs36Y_qibukJN8/edit?tab=t.0

- Note: There are two tabs, the link should take you to the one called “Blender Projection”. This is where the relevant information is.